

Effects of Integrating an AI Learning Companion into a Self-Regulated Learning 4L Framework on Elementary Mathematics Learning

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TALPer improves learning, but regulation explains who benefits



TALPer improves mathematics learning, but its strongest lesson is differentiated: students benefit when they regulate, question, monitor, and reflect with AI.

Problem

GenAI may scaffold or create dependence

Intervention

TALPer embedded in SRL 4L

Evidence

Outcome, process, and ENA networks

Takeaway

AI access is not enough

The GenAI promise comes with a dependence risk

Scaffold

Real-time feedback
Step-by-step guidance
Planning, monitoring, and reflection
Differentiated support

(Chiu, 2024; Wu et al., 2023)

Dependence

Passive answer-seeking
Procedural shortcutting
Reduced learner agency
Ability-level differences

(Jin et al., 2023; Lee & Low, 2024)

Grounded in Zimmerman's (2002) SRL cycle: forethought -> performance -> self-reflection

Research gap: whether this scaffold-dependence tension differs across learner ability levels remains underexplored.

Zimmerman's SRL Cycle x Ho's 4L Framework



Zimmerman (2002) SRL Cycle

Forethought

Goal setting - Prior-knowledge activation

Performance

Monitoring - Strategy - Peer adjustment

Self-Reflection

Error noticing - Conceptual consolidation

Ho (2014) SRL 4L Framework

Self-learning

Goal setting / prior activation

<-> Forethought

Co-learning

Monitoring & scaffolded feedback

<-> Performance

Mutual learning

Cross group strategy sharing

<-> Performance

Teacher-guided

Misconception fix & integration

<-> Self Reflection

The 4L phases map onto Zimmerman's cycle. We apply this mapping to design our SRL coding scheme and ENA dimensions.

TALPer: An AI Learning Companion

Original TALPer interface from the source deck



Student asks a mathematical question

TALPer responds through guided reasoning

Interaction logs become classroom evidence

This is the concrete system view behind the SRL 4L integration.

TALPer: an AI learning companion on TALP for elementary math problem solving (Kuo et al., 2026)

The study follows three evidence layers



RQ1 · Outcome

Does TALPer + SRL 4L improve learning outcomes compared with SRL 4L alone?

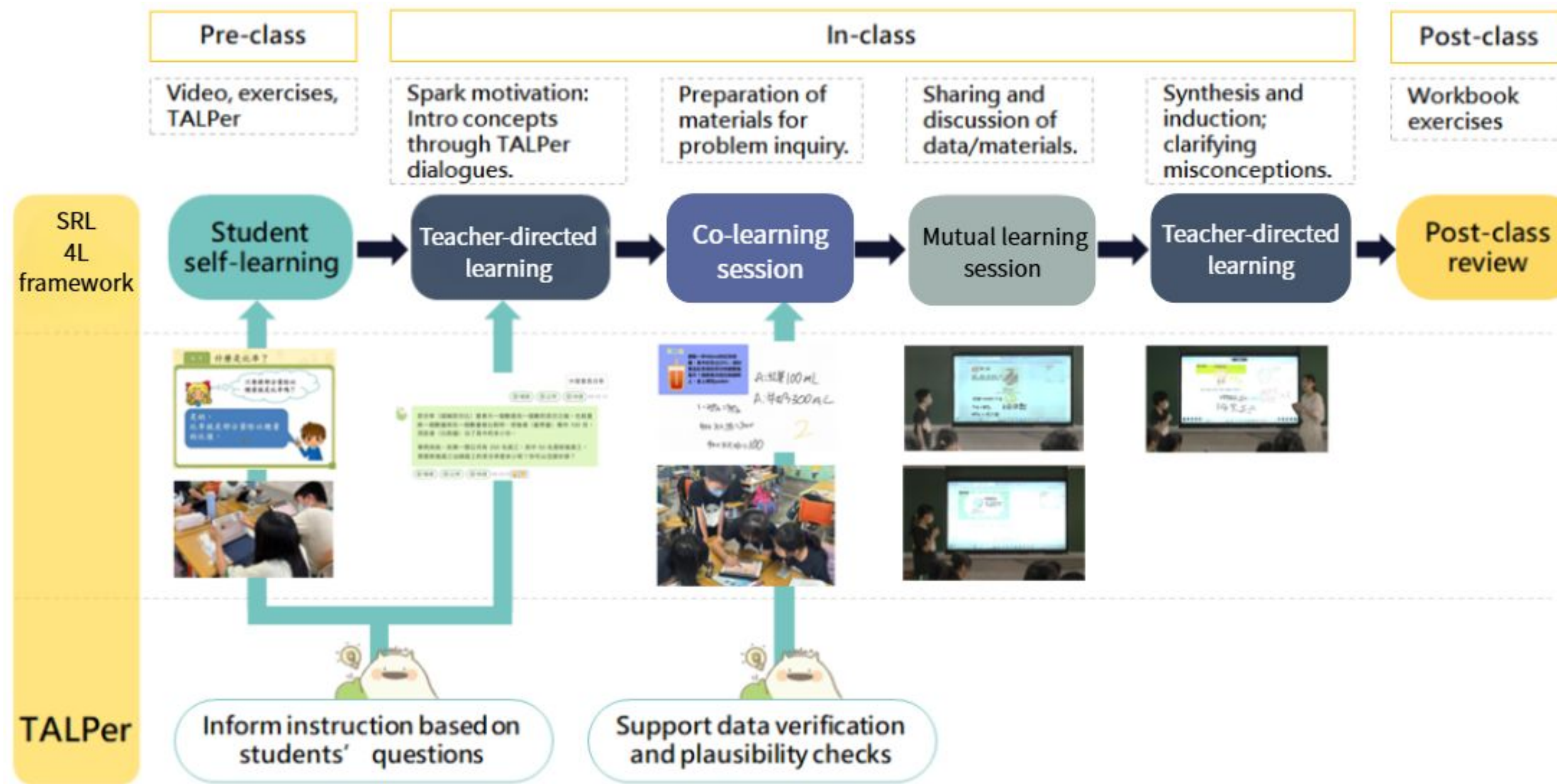
RQ2 · Process

Do high- and low-achieving students differ in SRL strategy use when TALPer is available?

RQ3 · Structure

Do their cognitive network structures differ in ENA representations?

The comparison isolates TALPer within the same 4L structure



Research Design and Participants



44

fifth-grade students

Public elementary school, Taiwan

21 / 23

experimental / control

No baseline difference ($p = .376$)

5

lessons on Ratios & Percentages

Delivered over 2 weeks

Comparison of the Two Groups

Experimental Group (n = 21)

- ▶ SRL 4L instruction + TALPer support throughout
- ▶ Pre-class: TALP videos + TALPer dialogues
- ▶ In-class: teachers leverage TALPer logs
- ▶ Co-/Mutual-/Teacher-guided learning all use TALPer

Control Group (n = 23)

- ▶ Identical SRL 4L instructional structure
- ▶ Only difference: no TALPer access
- ▶ Pre-test difference: not significant
- ▶ Isolates TALPer integration as the treatment

SRL Coding Framework and ENA Dimensions

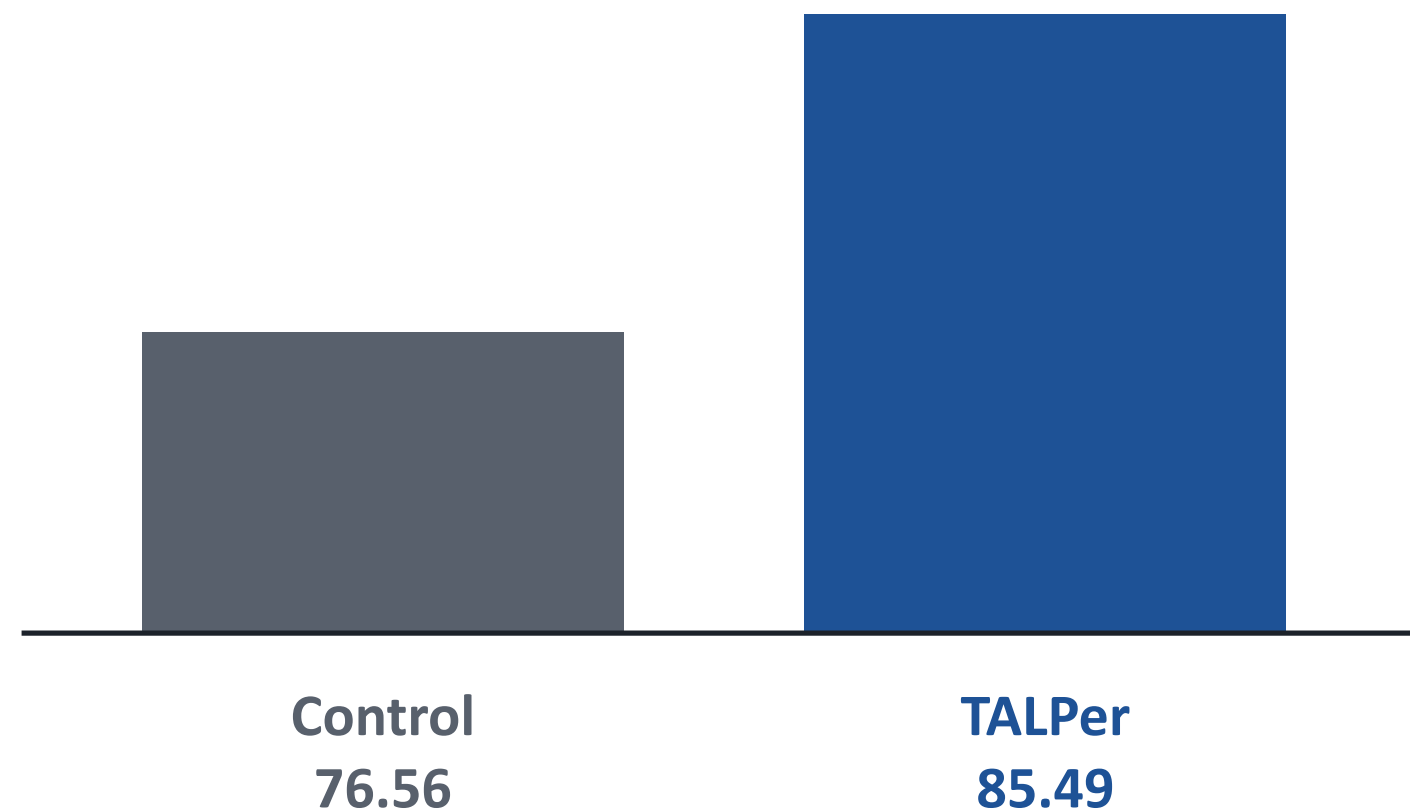


Main category	Sub-category	Code	Explanation
M Metacognition	Setting Goals	M.SG	Proactively sets or adjusts learning objectives or sequences.
	Selecting Strategies	M.SS	Selects or modifies learning strategies, modalities, or task forms.
	Monitoring & Evaluation	M.ME	Real-time awareness and evaluation of conceptual understanding.
	Reflection	M.R	Reviews problem solving to identify errors or revise strategies.
C Cognition	Conceptual Und. Ask	C.CUA	Asks questions about concepts, principles, meanings, or applications.
	Conceptual Und. Explain	C.CUE	Articulates or constructs conceptual understanding.
	Procedural Know. Ask	C.PKA	Asks about computational methods or problem solving steps.
	Procedural Know. Execute	C.PKE	Performs actual computations or formulates equations / answers.

ENA maps code co-occurrences into a 2D space -- X axis: relative cognitive tendencies; Y axis: structural complexity / diversity of the cognitive network.

TALPer-supported 4L produced higher adjusted post-test scores

Adjusted post-test means



ANCOVA result

$$F(1,41) = 4.71$$

$$p = .036$$

$$\text{eta squared} = .114$$

Pre-post gains: Control +14.45; TALPer +26.45 (both $p < .001$)

High- and low-achieving students used TALPer differently

High-achieving students

Goal setting
Conceptual questions
Monitoring and reflection
Iterative strategy use

Low-achieving students

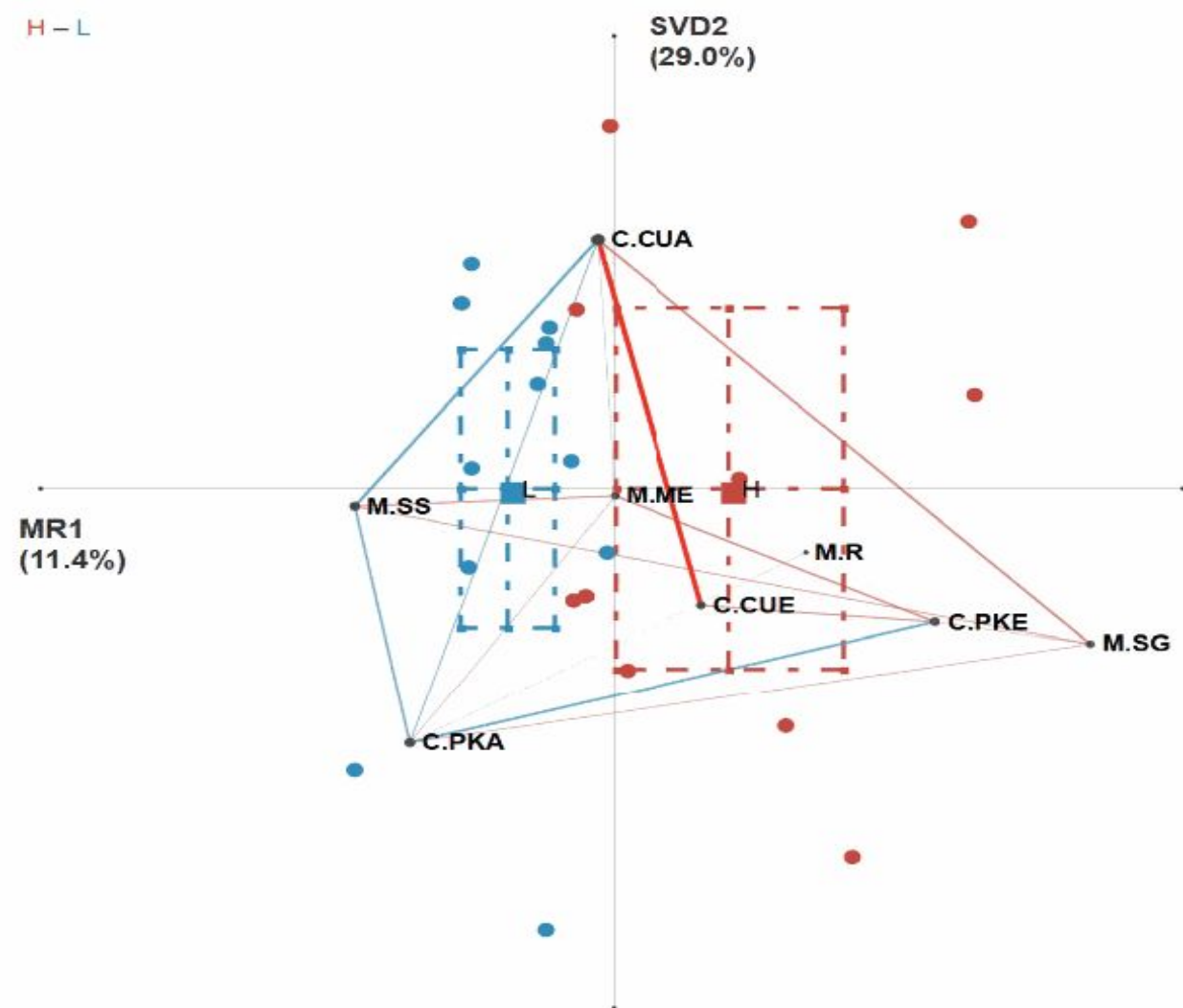
Strategy selection
Procedural asking
Answer-seeking flow
Limited reflection

$t(12.07) = 4.05 \cdot p < .001 \cdot \text{Cohen's } d = 1.83$

Experimental group only: high-achievers (n = 10) vs low-achievers (n = 11)

Comparison: ENA X-axis centroid difference between ability groups

ENA shows integrated versus fragmented cognitive networks



H: integrated, multidirectional
regulatory network

L: fragmented, procedural,
ask-execute pattern

Comparison of ENA network structures between high-achieving (H) and low-achieving (L) groups

Teachers need to teach students how to regulate with AI

TALPer's pedagogical effectiveness cannot be assumed from tool availability alone.

Guided questioning

Model how to ask conceptual questions, not only request answers.

Structured reflection

Use prompts that make students check, explain, and revise.

Dialogue modeling

Show examples of productive TALPer-mediated dialogue.

These supports are especially critical for low-achieving students who tend toward procedural dependence.

AI feedback should invite explanation, comparison, and reflection.

Take-home: AI scaffolding depends on regulatory awareness



An AI learning companion is not a silver bullet.

Its scaffolding effect depends on learners' regulatory awareness.

Thank you. Questions and discussion welcome.

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